

Regime-Based Extreme Price Formation in the GB Power Market: Evidence from Price Bands, Fuel Mix, Interconnectors and REMIT Events, 2023-2025

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Abstract

This study investigates the dynamics of extreme price formation within the Great Britain (GB) power market through a regime-based analysis of high frequency system data. Spanning the period from 2023 to 2025, the research integrates half-hourly system prices with a multi-dimensional dataset comprising fuel mix composition, Net Imbalance Volume (NIV), interconnector flows, and REMIT (Regulation on Wholesale Energy Market Integrity and Transparency) notifications. Rather than treating price spikes as isolated anomalies or the result of single market drivers, this analysis adopts a systematic perspective. It examines whether tail-risk events, both extreme peaks and negative pricing emerge from the complex interplay of intermittent wind, marginal gas dispatch, imbalance direction, and asset availability. This provides a robust framework for understanding price volatility within an increasingly decarbonised and interconnected grid.

This analysis uses Python-based data pipeline to evaluate Elexon BMRS system price and fuel mix data. Annual half-hourly datasets for 2023-2025 are used to classify market regimes, detect price events, and evaluate threshold robustness. Following an assessment of price-based transitions and case-study selection, a conditional spikeprobability framework determines how single, combined, and triple conditional stress conditions affect the likelihood of £250/MWh + events. REMIT notifications provide an external cross-check, ensuring asset-availability is interpreted as contextual evidence rather than direct causality.

Extreme GB power prices show strong regime dependency, with 2023 spikes reaching £1950/MWh during Q1 compound scarcity, particularly when high gas, low wind, and peak

demand coincided. Spike probability surged from 6.28% to over 34% during these combined conditions, demonstrating that multiple drivers, rather than single factors, dictate market risk. The 2024 market saw eased stress with a shift toward late-year physical scarcity, while early 2025 experienced both a £2900/MWh peak and negative price clusters tied to high wind. This evolution highlights a trend of increased volatility driven by specific, compounding market conditions.

The findings suggest that price formation in the GB power market is fundamentally driven by shifting regimes rather than isolated shocks. We see that extreme price movements are rarely random; instead, they emerge from the complex interaction of specific system variables. By shifting the focus away from single-driver outcomes and toward these compounding conditions, the proposed framework offers a more grounded and transparent lens for market monitoring. This approach provides a necessary foundation for risk analysis and future trading research as the power system becomes increasingly volatile and difficult to predict.

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1. Introduction

Extreme price movements in power markets rarely arise from a single factor. In modern electricity systems, where renewable generation, flexible thermal plant, interconnector flows and real-time balancing requirements interact, price formation is shaped by the relationship between physical system conditions and market timing. Previous studies have shown that electricity prices are extremely sensitive to demand, generation availability, fuel costs, renewable variability, and market design, making short-term price behaviour more volatile than in many other commodity markets (Bunn and Karakatsani, 2003; Weron, 2014; Knittel and Roberts, 2005).

This paper uses the Great Britain power market as a case study to examine how halfhourly system prices respond during periods of scarcity, surplus and operational stress between 2023 and 2025. The central argument of this study is that extreme prices should not be interpreted through individual variables such as low wind output, high gas generation, weak interconnector support or asset outages in isolation. Instead, upper-tail and lower-tail price outcomes are better understood through distinct market regimes. Factors such as imbalance direction, time of day, interconnector positioning and asset availability can all influence price formation, but their effect depends on the wider system state in which they occur.

This issue is increasingly important as renewable penetration rises. Wind generation can materially alter residual demand, changing the amount of flexible thermal generation, storage, imports or balancing actions required close to real time (Green and Vasilakos, 2010; Hirth, 2013; Cludius et al., 2014). Low wind output may increase reliance on gas generation and raise the probability of upper-tail prices, while high wind output during periods of weaker demand may increase exposure to low or negative prices. In this setting, price formation is not simply a question of whether one driver is present, but whether several drivers coincide.

The key research question is: **under what conditions do normally stressed market conditions escalate into extreme price events?** The analysis shows that single indicators have limited explanatory power on their own. For example, in the Q1 2023 stress regime, high gas generation alone was associated with a spike probability of around 6%. When low wind output coincided with the evening peak, the probability increased to around 20%. When positive imbalance was also present, the probability increased further to around 34%. This suggests that extreme price formation is driven less by individual drivers and more by the interaction between system conditions.

This study makes three main contributions. First, it provides a comparative assessment of GB power market price regimes across 2023, 2024 and 2025. Second, it applies a transparent Python-based analytical pipeline to identify regime fingerprints, price bands, threshold behaviour and single, combined and triple-condition spike probabilities. Third, it integrates REMIT notifications as an external cross-check, allowing asset-availability information to be assessed as contextual evidence rather than assuming direct one-outage causality.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature and market background. Section 3 describes the data and Python-based methodology. Section 4 presents the regime-level results, price-band analysis and conditional probability findings. Section 5 develops the case-study and REMIT crosscheck analysis. Section 6 discusses the implications for market monitoring, trading research and flexibility assets.

2. Literature Review and Market Background

2.1 Electricity price formation and volatility

Electricity prices differ from many other commodity prices because electricity must be balanced continuously and cannot be stored economically at large scale in the same way as

fuels such as gas, coal or oil. As a result, short-term electricity prices can respond sharply to changes in demand, generation availability, fuel costs, network conditions and market design. This has made electricity price volatility a major topic in power-market research, particularly in relation to forecasting, risk management and market monitoring (Bunn and Karakatsani, 2003; Knittel and Roberts, 2005; Weron, 2014).

In liberalised power markets, price formation is shaped not only by fundamental supply and demand conditions, but also by market rules, balancing arrangements and the timing of delivery. In the GB market, half-hourly system prices provide an important signal of balancing conditions because they reflect the cost of resolving system imbalances close to real time. Periods of high prices may occur when demand is strong, renewable output is low, gas-fired generation is heavily relied upon, or interconnector support is limited. By contrast, low or negative prices are more likely when renewable output is high, demand is weaker, thermal generation is reduced, and the system is exposed to negative imbalance.

2.2 Renewable variability, residual demand and price tails

A major theme in the literature is that increasing renewable generation can change the distribution of electricity prices. Wind and solar generation have low short-run marginal costs, which can reduce average wholesale prices during periods of high renewable output. However, variable renewable generation can also increase price dispersion because the system must respond to changing residual demand, forecast error and flexibility requirements (Green and Vasilakos, 2010; Hirth, 2013; Cludius et al., 2014).

For the GB system, wind generation is particularly important because it can materially change the amount of demand that must be served by gas generation, imports, storage and other flexible resources. When wind output is low, the system may require higher gas generation and more balancing actions, increasing the probability of upper-tail price events. When wind output is high, especially during lower-demand periods, surplus conditions can increase exposure to low or negative prices. This means that both positive and negative price extremes can be linked to renewable variability, but through different system regimes.

2.3 Gas generation, scarcity and upper-tails prices

Gas-fired generation remains an important source of flexibility in the GB power system. During periods of low wind output or high demand, gas generation often plays a larger role in meeting residual demand and maintaining system balance. Previous work on electricity market fundamentals has shown that thermal generation, fuel prices and scarcity conditions can strongly influence short-term price outcomes, particularly when system margins are tight (Newbery, 2005; Bunn and Karakatsani, 2003; Weron, 2014).

However, high gas generation alone does not necessarily imply an extreme price event. Its effect depends on the wider market state, including wind output, imbalance direction, interconnector flows and time of day. This is central to the regime-based approach used in this paper. For example, this study's threshold analysis shows that periods above £300/MWh

in 2025 were associated with average gas generation of around 22.8 GW, average wind output of around 3.0 GW, and low wind below 8 GW in 100% of those periods. This suggests that upper-tail prices were most clearly associated with combined physical conditions rather than one isolated driver

2.4 Negative prices and surplus conditions

Negative electricity prices have become more common in power systems with high renewable penetration, particularly when low marginal cost generation coincides with weak demand, limited flexibility or network constraints. Existing research links negative prices to surplus renewable generation, inflexible generation, policy incentives and limited demand-side response (Genoese et al., 2010; Cludius et al., 2014; Hirth, 2013).

In the GB context, negative prices can occur when high wind output and negative imbalance create downward pressure on system prices. These lower-tail events should not be treated as the opposite of scarcity spikes in a simple linear way. Instead, they represent a different regime of price formation, where surplus conditions, reduced thermal demand and balancing requirements push prices downward. This paper therefore analyses both upper-tail and lower-tail price behaviour to show how different combinations of system conditions create different forms of price stress.

2.5 Interconnectors and cross-border market conditions

Interconnectors are another important part of GB price formation. Cross-border flows can provide support during tight domestic conditions by allowing imports from neighbouring markets. However, interconnectors do not always act as a stabilising factor. Their contribution depends on market spreads, availability, flow direction and conditions in connected systems. In some periods, reduced import support or export pressure can reinforce domestic scarcity or surplus conditions (Newbery, 2005; ACER, 2023; ENTSO-E, 2023).

For this reason, this study treats interconnector flows as regime dependent.

Interconnectors are analysed alongside fuel mix, imbalance and time bands rather than as an isolated explanatory variable. This allows the paper to examine whether interconnector conditions reinforce or weaken price stress under different market regimes.

2.6 REMIT notifications and asset availability

Asset availability is another important part of short-term price formation. In European power markets, REMIT was introduced to improve transparency and reduce market abuse by requiring market participants to publish inside information, including information about generation, consumption and transmission unavailability (ACER, 2023; Ofgem, 2024). REMIT notifications therefore provide useful external evidence on asset availability around price events.

However, REMIT data must be interpreted carefully. A notification showing unavailable capacity does not automatically mean that the outage caused a price spike. The impact of an outage depends on the wider system condition at the time. For example, a plant outage during a low-demand, high-wind period may have limited price effect, while a similar outage during low wind, high demand and positive imbalance may be more material. This paper therefore uses REMIT notifications as an external cross-check rather than as a direct causal explanation for every price movement.

2.7 Research gap and contribution

The existing literature provides strong evidence that electricity prices are affected by renewable variability, fuel costs, demand, balancing conditions, interconnection and asset availability. However, many studies examine these drivers separately or focus mainly on forecasting performance. Less attention is given to how different combinations of system conditions create distinct price regimes and how those regimes affect the probability of extreme price outcomes.

This paper addresses that gap by developing a regime-based empirical framework for the GB power market between 2023 and 2025. The framework combines price-band analysis, seasonal regime comparison, threshold robustness, single-condition and multi-condition spike probabilities, case-study analysis and REMIT cross-checking. This allows the study to assess not only whether extreme prices occurred, but also what system conditions were present when normal market stress escalated into upper-tail or lower-tail price behaviour.

3. Data and Methodology

3.1 Data Sources

This study uses publicly available Great Britain electricity market data to examine how system prices behave under different market conditions between 2023 and 2025. The main data source is Elexon BMRS, which provides settlement-period data for the GB balancing and wholesale electricity market (Elexon Limited, 2026). The analysis focuses on half-hourly observations because this is the resolution at which system price, fuel mix, imbalance conditions and interconnector flows can be linked most directly.

The core price variable used in the study is the GB system price. This is used to identify normal price periods, elevated prices, positive spike periods and negative-price periods. Alongside system price, the analysis uses fuel mix data to capture changes in gas generation, wind generation, nuclear output, biomass, hydro, pumped storage, coal, oil, other generation and interconnector flows. These variables are used to assess whether price stress is associated with changes in the physical structure of supply.

Net imbalance volume is also included because the direction and scale of system imbalance can influence final balancing prices. Positive imbalance, negative imbalance and the size of the imbalance are treated as part of the wider market state rather than as isolated explanatory variables. This is important because the same level of wind or gas generation may have different price effects depending on whether the system is short, long or close to balance.

For the 2025 case-study analysis, REMIT notifications are used as an external crosscheck. These notifications provide information about generation, consumption and transmission availability events around selected price windows. In this paper, REMIT data is not treated as automatic evidence that a specific outage caused a price movement. Instead, it is used to assess whether asset-availability conditions were present around major price events and whether those conditions were case-specific or part of repeated background availability patterns.

The analysis was implemented using a Python-based workflow. The scripts and generated analytical tables were organised in a version-controlled GitHub repository to support reproducibility and auditability of the analysis. The repository is used to maintain the data-processing pipeline, validation outputs, regime tables, price-band summaries, conditional probability outputs and case-study results.

After cleaning and merging the datasets, the final 2025 dataset contains 17,520 halfhourly rows, covering the full period from 1 January 2025 00:00 to 31 December 2025 23:30, with no missing or duplicate timestamps. The validation summary identifies 42 periods above £250/MWh, 27 periods above £300/MWh, 26 near-spike periods between £200/MWh and £250/MWh, and 1,121 negative-price periods in 2025. These validated datasets form the basis for the price-band, regime, threshold and case-study analysis developed in the following sections.

3.2 Data Construction and validation

The analysis was built around yearly half-hourly datasets for 2023, 2024 and 2025. For each year, system price data was combined with fuel mix, net imbalance volume and interconnector variables using settlement-period timestamps as the common time index. This allowed each half-hourly price observation to be linked with the physical market conditions present at the same time, including gas generation, wind generation, interconnector flows and imbalance direction.

The dataset construction process followed a consistent sequence. First, raw Elexon BMRS data was collected for the relevant year (Elexon Limited, 2026). Second, timestamp fields were standardised so that observations could be aligned on a halfhourly basis. Third, generation variables were cleaned and arranged into a common structure across all three years. Finally, system price and imbalance data were merged with the fuel mix dataset to create a single analytical table for each year. This approach ensured that price behaviour could be assessed against the wider system state rather than against price data alone.

A validation step was then carried out to check whether the annual datasets were complete and suitable for analysis. This included checking the number of rows, the first and last timestamp, missing timestamps, duplicate timestamps and the presence of key price-event categories. The final annual datasets contained 17,512 rows for 2023, 17,568 rows for 2024 and 17,520 rows for 2025. The slightly different row counts reflect the structure and completeness of the source data for each year, including leap-year effects in 2024. The annual summary also confirmed that the maximum observed system price was £1,950/MWh in 2023, £669.212/MWh in 2024 and £2,900/MWh in 2025. Negative-price periods were also present in all three years, with 849 periods in 2023, 1,073 periods in 2024 and 1,121 periods in 2025.

The validation process was important because the study relies on comparing price behaviour across years and regimes. If missing or duplicate timestamps were left unresolved, price-event counts and conditional probabilities could be distorted. For this reason, the analysis did not move directly from raw data to results. Each dataset was first checked for coverage, consistency and basic price-event classification before being used in the regime, threshold and case-study analysis.

Once validated, the annual datasets were used to produce a series of derived tables. These included annual market summaries, monthly regime summaries, price-band driver comparisons, threshold robustness checks, conditional spike probability tables and selected case-study windows. These outputs formed the analytical foundation of the paper. The purpose of constructing the dataset in this way was not only to identify whether extreme prices occurred, but also to examine what type of system conditions were present when those price events developed.

3.3 Price – band classification

To understand how market conditions change as prices move from normal levels into stressed or extreme levels, the analysis classifies each half-hourly observation into a set of price bands. This step is important because the study is not only concerned with whether a price spike occurred, but also with what begins to change before the market reaches an extreme price outcome. In other words, the price-band method helps show how the system behaves at different price levels.

The bands used in the analysis separate negative prices, normal price periods, elevated prices, near-spike periods and extreme positive price periods. Negative prices are treated separately because they usually reflect a different type of system condition, often linked to high renewable output, negative imbalance and lower thermal demand. Positive price bands are then grouped progressively, moving from normal prices through to higher stress levels. The main positive bands used in the analysis are £0–100/MWh, £100–150/MWh, £150–200/MWh, £200–250/MWh, £250–300/MWh and £300/MWh and above.

This classification allows the analysis to compare average market conditions across price levels. For each price band, the study calculates average system price, average imbalance,

average wind generation, average gas generation and average interconnector flow. It also calculates the share of periods in each band where certain stress conditions are present, such as wind below selected thresholds, gas generation above selected thresholds, positive or negative imbalance, low interconnector support and evening peak timing.

The purpose of this method is to avoid treating extreme prices as sudden unexplained events. Instead, it allows price formation to be observed as a transition across bands. For example, in 2025, the normal £0–100/MWh band had average gas generation of around 8.3 GW and average wind output of around 8.1 GW. By contrast, the £300/MWh and above band had average gas generation of around 22.8 GW and average wind output of around 3.0 GW. In the same £300/MWh and above band, wind output was below 8 GW in all periods, gas generation exceeded 15 GW in around 85% of periods, and evening peak timing appeared in around 63% of periods.

The price-band classification therefore gives the paper a structured way to answer one of its main questions: at what price level do the underlying system drivers begin to change? It shows that the movement from normal prices to extreme prices is not only a price movement, but also a change in the physical and balancing state of the system.

This provides the foundation for the later threshold, conditional probability and case study analysis.

3.4 Regime classification

After the price bands were created, the analysis grouped each year into wider market regimes. This was done to capture the fact that the same driver can have a different price effect depending on the surrounding market context. For example, low wind output during a quiet surplus period may not produce the same price response as low wind output during an evening peak with high gas generation and positive imbalance. The regime classification therefore provides a way to compare periods with similar physical and balancing conditions, rather than treating the year as one uniform market environment.

The regimes were defined using a combination of calendar timing, observed price behaviour and system-condition fingerprints. Each regime was assessed using average price, maximum and minimum price, imbalance direction, wind generation, gas generation, interconnector flows, negative-price frequency and the number of periods exceeding selected price thresholds. This allowed the study to identify whether a period was mainly scarcity-led, surplus-led, quiet, transitional or mixed-tail in character.

For 2023, the analysis identified a Q1 compound scarcity architecture. This period showed concentrated upper-tail activity, including the main 2023 price spike of £1,950/MWh. The Q1 regime was not simply a “high price” period; it combined elevated price frequency, evening amplification and interaction between gas, wind and imbalance conditions. By contrast, the April–September period was quieter and more surplus-leaning, showing that low wind alone

was not enough to consistently produce price spikes when the wider market regime was less stressed.

For 2024, the annual structure was different. Q1 was relatively quiet, unlike 2023.

Upper-tail stress began to return later in the year, especially during the October– November transition and December physical scarcity regimes. The December regime contained the main 2024 stress events, with maximum prices above £500/MWh and stronger evidence of low wind and high gas conditions. This made 2024 a useful comparison year because it showed that scarcity did not appear in the same seasonal position as in 2023.

For 2025, the regime structure was more mixed. January formed the main upper-tail shock regime, containing the largest observed price event in the study period at £2,900/MWh and the highest concentration of £300/MWh and above periods. Later regimes showed different forms of stress. April–May and July–September was more surplus-sensitive, with greater negative-price exposure, while June and October showed mixed-tail behaviour, where negative-price risk and renewed positive spike risk both appeared within the same broader annual architecture.

This regime classification is central to the paper because it allows the analysis to compare not only prices, but also the market conditions behind those prices. Across the three-year period, the results show that extreme price formation was not evenly distributed across time. It was concentrated in specific regimes: Q1 compound scarcity in 2023, late-year physical scarcity in 2024, and a mixed scarcity-surplus architecture in 2025. This supports the paper’s main argument that GB price tails are regimedependent outcomes rather than random isolated events.

3.5 Threshold and spike-event definition

A key methodological issue in any study of extreme electricity prices is how to define a “spike”. A single threshold can be useful, but it can also hide important information about how price stress develops. For this reason, the analysis does not rely only on one fixed cut-off. Instead, it tests several price thresholds, including £100/MWh, £150/MWh, £200/MWh, £250/MWh and £300/MWh. This allows the study to examine whether the same system drivers remain important as the definition of price stress becomes more severe.

The threshold approach serves two purposes. First, it separates moderate price elevation from genuine upper-tail behaviour. A price above £100/MWh may indicate a tighter market than normal, but it does not necessarily represent an extreme event. By contrast, prices above £250/MWh or £300/MWh are more useful for identifying uppertail stress, especially when they occur alongside low wind output, high gas generation, positive imbalance or weak interconnector support. Second, testing multiple thresholds helps check the robustness of the results. If similar drivers appear across higher thresholds, this gives stronger evidence that those conditions are genuinely linked to price-tail formation.

The analysis shows that the composition of the market changes as the price threshold increases. In 2023, periods above £250/MWh were strongly associated with low wind and high gas generation. Around 83% of these periods had wind output below 8 GW, while around 74% had gas generation above 15 GW. At the £300/MWh threshold, the low-wind share increased to around 90%, and the share of evening-peak periods increased to around 86%. This suggests that the most extreme 2023 prices were closely linked to a combination of physical scarcity and timing effects.

The same pattern is visible in 2024, although the number of extreme price periods was much smaller. For prices above £250/MWh and £300/MWh, all periods occurred when wind was below 8 GW and gas generation was above 15 GW. This supports the interpretation that 2024 upper-tail stress was more limited but more physically concentrated when it did occur.

In 2025, threshold behaviour became more severe again. Periods above £250/MWh had average gas generation of around 22.2 GW and average wind output of around 3.0 GW.

At the £300/MWh threshold, average gas generation increased further to around 22.8 GW, while all periods remained below the 8 GW wind threshold. Around 63% of £300/MWh and above periods also occurred during the evening peak. This confirms that the largest 2025 price events were not isolated price movements but were linked to a wider scarcity fingerprint involving low wind, high gas generation and market timing.

Negative-price events are treated separately from positive spikes. The reason is that negative prices represent a different form of price stress. They are usually associated with surplus conditions, high renewable output, negative imbalance, lower thermal demand or limited flexibility. Therefore, the paper distinguishes between upper-tail spike analysis and lower-tail negative-price analysis, rather than combining them into one general category of “extreme prices”.

Overall, the threshold method allows the study to test whether its findings depend on the chosen spike definition. The results show that as the threshold rises, the market fingerprint becomes more concentrated around low wind, high gas generation, imbalance stress and time-band effects. This supports the use of a regime-based approach, because the most extreme prices appear when several system conditions align rather than when a single driver is present on its own.

3.6 Conditional spike-probability framework

To test whether extreme prices were linked to individual drivers or to combinations of system conditions, the study uses a conditional spike-probability framework. This method calculates how often a price spike occurred when a selected condition, or combination of

conditions, was present. The purpose is not to build a predictive model at this stage, but to measure how the probability of a spike changes as different stress conditions are layered together.

The basic calculation is:

Spike probability given condition = number of spike periods within the condition / total periods meeting the condition

In this study, a spike is mainly treated as a period where system price exceeds the selected upper-tail threshold, particularly £250/MWh. Conditions are then tested individually and in combination. The single-condition tests include variables such as positive imbalance above selected thresholds, wind generation below selected thresholds and gas generation above selected thresholds. These tests help identify whether a single driver appears to be associated with higher spike risk.

However, the main value of the framework comes from the combined-condition and triple-condition tests. Combined-condition tests examine whether spike probability increases when two drivers occur together, such as low wind and high gas generation, positive imbalance and high gas generation, or low wind and evening peak timing. Triple-condition tests then examine whether spike risk rises further when three conditions occur at the same time, such as positive imbalance, low wind and evening peak timing.

This approach is useful because power-market stress is rarely caused by one variable acting alone. A low-wind period may not produce an extreme price if demand is weak, imbalance is low or interconnector support is strong. Similarly, high gas generation may reflect normal seasonal demand rather than scarcity if it does not coincide with other stress conditions. By comparing single, combined and triple-condition probabilities, the analysis can show whether price spikes become more likely when several system pressures align.

The Q1 2023 stress regime provides a clear example of this method. In the singlecondition tests, gas generation above 15 GW was associated with 82 spike periods out of 1,306 periods meeting the condition, giving a spike probability of 6.28%. Positive imbalance above 150 MW had a similar spike probability of 6.07%, while wind below 8 GW had a spike probability of 4.81%. These results suggest that individual drivers were relevant, but not sufficient on their own to explain the highest-risk periods.

The combined-condition tests show a stronger pattern. When positive imbalance above 150 MW coincided with gas generation above 15 GW, the spike probability increased to 13.78%. When positive imbalance coincided with the 16:00–19:00 evening window, it increased to 17.45%. The highest two-condition result was low wind below 8 GW combined with the evening window, where 63 spike periods occurred out of 318 qualifying periods, giving a spike probability of 19.81%.

The triple-condition tests show the clearest escalation. When positive imbalance above

150 MW, low wind below 8 GW and the 16:00–19:00 evening window occurred together, 47 of 138 qualifying periods were spike periods. This produced a conditional spike probability of 34.06%. This was much higher than the single-condition results and supports the paper’s argument that price-tail formation is driven by interacting system conditions rather than by one driver in isolation.

The conditional probability framework is therefore used as a bridge between descriptive market analysis and case-study interpretation. It quantifies how spike risk changes as drivers combine, while remaining transparent and easy to interpret. This is important for the purpose of the paper because the objective is not only to identify extreme prices, but to explain the market conditions under which normal stress escalates into uppertail price behaviour.

3.7 Case-study selection

The case-study stage was added to connect the statistical results with specific market events. While the price-band, regime and conditional probability analysis shows the broad pattern of price formation, selected cases allow the analysis to examine how those patterns appeared in real half-hourly market windows. The case studies were therefore not chosen simply because they had high prices. They were selected because they represented important regime behaviours within the wider three-year dataset.

For 2023 and 2024, the case studies were selected to provide historical benchmarks for the regime-based framework. The 2023 case represents the Q1 compound scarcity regime, where upper-tail price stress was concentrated early in the year and reached a maximum system price of £1,950/MWh. The 2024 cases represent a different annual structure: a late-year transition event in October, followed by December physical scarcity events. These cases are used to show that price stress did not appear in the same seasonal pattern every year. In 2023, the main stress was concentrated in Q1, while in 2024 it returned later in the year, especially during the October–December period.

For 2025, a wider set of case studies was selected because the year contained both severe positive spike behaviour and repeated negative-price clusters. The selected 2025 cases include the January extreme spike event reaching £2,900/MWh, further January scarcity events, June and October positive-spike cases, a July near-spike case and several negative-price clusters across March, April, June and September. This selection allows the paper to examine both upper-tail and lower-tail price formation within the same annual framework.

Each 2025 case was defined using an event window around the relevant price behaviour. These windows were then summarised using maximum price, minimum price, average price, dominant time band, regime group, average imbalance, average wind generation, average gas generation and average interconnector flow. This gave each case a consistent internal market fingerprint. For example, the January 2025 extreme spike case had a maximum price of £2,900/MWh, average wind output of around 2.8 GW and average gas generation above

24 GW within the selected window. By contrast, the September 2025 negative-price case was linked to very high wind output, negative imbalance and lower-tail price behaviour.

The purpose of using case studies is not to replace the wider statistical analysis, but to test whether the same regime logic appears in specific events. The selected cases therefore act as a bridge between the aggregated results and the REMIT cross-check. They show whether the system conditions identified in the price-band and conditional probability analysis can also be observed during major real-world price events.

The final case-study table contains 16 cases: one 2023 case, three 2024 cases and twelve 2025 cases. The 2025 cases are more detailed because they were reassessed using the REMIT overlay, while the 2023 and 2024 cases are used mainly as historical regime benchmarks. This distinction is important, because the paper does not claim the same level of REMIT-based evidence for all years. Instead, the 2023 and 2024 cases establish the historical price-regime context, and the 2025 cases extend the analysis with asset-availability cross-checking.

3.8 REMIT cross-checked method

The final stage of the methodology links selected 2025 price-event windows with REMIT notifications (Elexon Limited, 2026). This step was included because system price, fuel mix and imbalance data describe the internal market state, but they do not fully show whether asset-availability events were present around the same time. REMIT data therefore provides an external layer of evidence that can be compared with the internal market fingerprint of each case.

For each 2025 case-study window, REMIT events were collected within the relevant external check window. The initial REMIT extraction included generation, consumption and transmission unavailability events. These events were then filtered to focus on materially relevant notifications, with particular attention to event type, unavailable capacity, affected asset and whether the event overlapped with the selected price window. The purpose was to avoid treating every notification as equally important.

A further classification step was applied to distinguish between case-specific availability evidence and repeated background availability conditions. This distinction is important. Some assets appeared repeatedly across several 2025 case windows, meaning that their presence in a REMIT search did not necessarily explain one specific price event. In those cases, the REMIT evidence was interpreted as repeated external availability background rather than direct single-outage causality.

The final 2025 REMIT classification shows that all twelve 2025 case windows were best classified as **repeated background external context**. This means that REMIT evidence showed material asset unavailability during the selected case windows, but the dominant assets often recurred across multiple cases. For example, the January 2025 extreme spike case included repeated assets such as NGIFA, SSETGSL, KGL, BEGL001 and DRAX, with NGVLL

identified as a case-specific asset. The October 2025 cases also included repeated assets such as NGIFA, CENLANG, ICHPLLP, BEGL001, DRAX and CECL, with ELECL1 appearing as a case-specific asset in some windows.

This approach makes the REMIT analysis more conservative. The paper does not claim that a single outage caused each price spike. Instead, REMIT notifications are used to test whether external availability stress was present and whether it reinforced the regime interpretation. The primary explanation remains the internal market state: wind output, gas generation, imbalance direction, interconnector support, time band and regime group. REMIT evidence is then used as a cross-check to show whether asset availability formed part of the wider context.

This method is especially important for the paper's central argument. If REMIT events are treated too simply, the analysis risks becoming a list of outages near price spikes. By separating repeated background context from case-specific evidence, the study avoids that problem. The result is a more careful interpretation: extreme GB price events appear to be regime-led, with REMIT notifications acting as external context rather than proof of simple one-outage causality.

4. Results and Analysis

4.1 Annual market overview: 2023-2025

The annual comparison shows that the GB power market did not follow the same price structure across 2023, 2024 and 2025. Each year had a different balance between average price level, upper-tail spike exposure and negative-price frequency. This supports the argument that extreme price formation should be analysed through yearly and seasonal regimes rather than through one fixed market pattern.

In 2023, the average system price was £94.55/MWh, the highest annual average across the three-year period. The year also contained the main 2023 upper-tail event, with a maximum system price of £1,950/MWh. There were 120 half-hourly periods at or above £250/MWh and 21 periods at or above £300/MWh. This indicates that 2023 had a relatively strong upper-tail structure, even though negative prices were also present. The annual dataset recorded 849 negative-price periods, equal to around 4.85% of all halfhourly periods.

The 2024 market structure was different. The average system price fell to £71.17/MWh, and the maximum observed price was lower, at £669.21/MWh. The number of highprice periods also declined sharply. Only 24 periods reached or exceeded £250/MWh, while 17 periods reached or exceeded £300/MWh. This suggests that 2024 was generally less exposed to sustained upper-tail stress than 2023. However, the presence of late-year price events shows that the year was not entirely quiet; instead, stress appeared more selectively, especially during the October–December period.

In 2025, the average system price increased to £80.56/MWh, but the more important feature was the wider tail behaviour. The maximum system price reached £2,900/MWh, the highest value in the full 2023–2025 dataset. There were 42 periods at or above £250/MWh and 27 periods at or above £300/MWh. At the same time, 2025 also recorded 1,121 negative-price periods, the highest count across the three years, equal to around 6.40% of all half-hourly periods. This shows that 2025 was not simply a high-price year.

It was a mixed-tail year, containing both severe positive spikes and frequent negative-price exposure.

Across the three years, the annual summary therefore shows a changing price architecture. The 2023 market was characterised by a stronger average price level and concentrated upper-tail stress. The 2024 market was lower-priced on average, with more selective late-year scarcity events. The 2025 market combined the most severe positive price event with the highest number of negative-price periods. This supports the paper's central argument that GB price extremes are regime-dependent and that upper-tail and lower-tail behaviour can coexist within the same market year.

4.2 Regime-level comparison

The annual summary shows how the three years differed overall, but it does not fully explain when and why price stress appeared. To address this, the analysis separates each year into regime groups. This gives a clearer view of how extreme prices were concentrated in parts of the year rather than being evenly distributed across all settlement periods.

In 2023, the main stress regime was concentrated in Q1. The Q1 compound scarcity regime had an average price of £132.54/MWh and reached a maximum price of £1,950/MWh. It contained 111 periods at or above £250/MWh and 18 periods at or above £300/MWh. This was much higher than the rest of the year, showing that 2023 upper-tail stress was not a general annual condition, but a concentrated Q1 event structure. The April–September period had only one £250/MWh period, despite containing many low-wind periods, which suggests that low wind alone was not enough to create frequent spikes without the wider scarcity regime.

In 2024, the regime structure was different. Q1 2024 was relatively quiet and did not produce any £250/MWh or £300/MWh periods. The main upper-tail behaviour returned later in the year, particularly in the October–November transition and December physical scarcity regimes. The December 2024 regime reached a maximum price of £590.45/MWh and contained 14 periods at or above £250/MWh and 11 periods at or above £300/MWh. This supports the interpretation that 2024 stress was more selective and more late-year focused than 2023.

The 2025 structure was more complex. January formed the main upper-tail shock regime, with an average price of £123.87/MWh, a maximum price of £2,900/MWh and 19 periods at or above £300/MWh. However, later parts of 2025 showed a different type of stress. April–

May and July–September was more surplus-sensitive, with increased negative-price exposure and weak positive spike activity. June and October showed mixed-tail behaviour, where negative-price periods and renewed positive spike risk appeared within the same broader regime. For example, the June mixed-tail regime recorded 227 negative-price periods while also containing six periods at or above £250/MWh and four periods at or above £300/MWh.

This regime-level comparison supports one of the central findings of the paper: extreme price formation in the GB market is not evenly distributed and cannot be explained by annual averages alone. The same market can move between scarcity-led, surplus-led and mixed-tail conditions within a single year. In 2023, upper-tail stress was concentrated in Q1. In 2024, it shifted towards late-year physical scarcity. In 2025, the market showed both severe upper-tail and lower-tail behaviour, with January scarcity, summer surplus exposure and autumn mixed-transition stress all appearing in the same year.

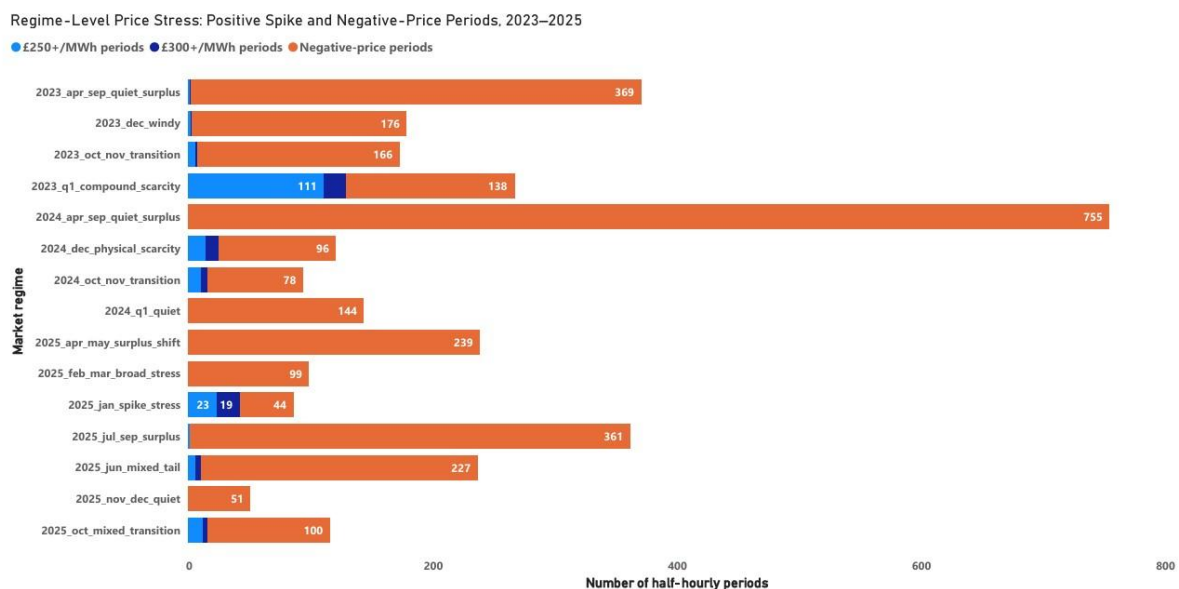


Figure 4.2a Regime-Level Price Stress: Positive Spike and Negative-Price Periods, 2023–2025

The Figure 4.2 compares the number of half-hourly periods above £250/MWh, above £300/MWh and below £0/MWh across each market regime. It shows that upper-tail stress was concentrated in scarcity regimes, while negative-price exposure was more visible in surplus and mixed-tail regimes.

Regime Physical Driver Fingerprint: Wind, Gas and Interconnector Conditions, 2023–2025

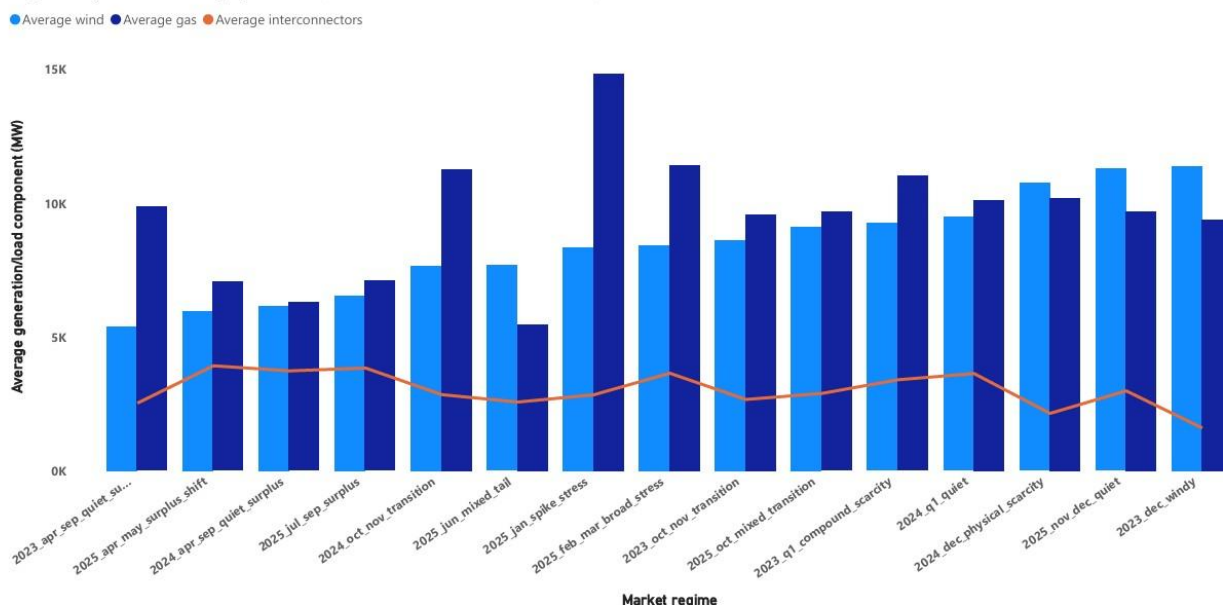


Figure 4.2b Regime physical driver fingerprint across wind, gas and interconnector conditions, 2023–2025.

The figure compares average wind generation, gas generation and interconnector flow across market regimes. Scarcity-led regimes show stronger gas reliance and lower wind conditions, while surplus or mixed-tail regimes show a different balance between wind output, gas generation and interconnector support.

Regime-level market fingerprint comparison

Year	Market regime	Avg price (£/MWh)	Max price (£/MWh)	Negative-price periods	£250+/MWh periods	£300+/MWh periods
2023	2023_apr_sep_quiet_surplus	81.51	370.00	369	1	1
2023	2023_dec_windy	69.02	300.00	176	2	1
2023	2023_oct_nov_transition	90.67	310.00	166	6	1
2023	2023_q1_compound_scarcity	132.54	1,950.00	138	111	18
2024	2024_apr_sep_quiet_surplus	66.04	200.00	755	0	0
2024	2024_dec_physical_scarcity	86.26	590.45	96	14	11
2024	2024_oct_nov_transition	87.59	669.21	78	10	6
2024	2024_q1_quiet	65.33	177.71	144	0	0
2025	2025_apr_may_surplus_shift	72.08	204.11	239	0	0
2025	2025_feb_mar_broad_stress	93.90	207.40	99	0	0
2025	2025_jan_spike_stress	123.87	2,900.00	44	23	19
2025	2025_jul_sep_surplus	71.59	263.58	361	1	0
2025	2025_jun_mixed_tail	69.21	423.85	227	6	4
2025	2025_nov_dec_quiet	75.63	247.31	51	0	0
2025	2025_oct_mixed_transition	75.89	487.00	100	12	4

Table 4.2a Regime-level price outcome summary, 2023–2025

The regime-level results show that price stress was unevenly distributed across the study period. Figure 4.2a shows that upper-tail price events were concentrated in a small number of regimes rather than spread evenly across the full dataset. The 2023 Q1 compound scarcity regime contained the largest number of £250/MWh and above periods in 2023, while 2024 price stress was more concentrated in the October– December period. In 2025, the January

spike-stress regime contained the strongest positive price shock, but later regimes also showed evidence of mixed-tail behaviour.

The same figure also shows that negative-price exposure followed a different pattern from positive spike exposure. Negative-price periods were more visible during surplus or mixed-tail regimes, particularly during the 2024 April–September quiet surplus period and several 2025 surplus-sensitive regimes. This supports the view that upper-tail and lower-tail price events should not be treated as one general category of “extreme prices”. They represent different system conditions.

Figure 4.2b helps explain the physical driver differences behind these regimes. Scarcity-led regimes generally show stronger reliance on gas generation and lower wind conditions, while surplus or lower-price regimes are more closely linked to higher wind output and weaker thermal generation requirements. Interconnector flows vary across regimes, suggesting that interconnector support should be treated as part of the wider market state rather than as a single stabilising variable.

Table 4.2 summarises these regime-level price outcomes. Taken together, the results show that GB price formation between 2023 and 2025 was regime dependent. The main upper-tail events occurred when scarcity conditions, timing effects and physical system drivers aligned, while negative-price periods were more associated with surplus-sensitive regimes. This supports the paper’s wider argument that extreme prices are better understood through interacting system conditions rather than isolated variables.

4.3 Price – band driver comparison

The regime analysis shows when price stress was concentrated, but it does not fully show how market conditions changed as prices moved from normal levels into extreme territory. For this reason, the analysis also compares system drivers across price bands. This helps identify whether physical and balancing conditions begin to change gradually as prices rise, or whether the strongest driver changes only appear in the upper-tail bands.

The price-band results show a clear difference between normal-price periods, elevated-price periods and extreme-price periods. Across the three years, higher positive price bands were generally associated with lower wind output, higher gas generation and a greater share of positive imbalance periods. This pattern was strongest in the upper-tail bands above £250/MWh and £300/MWh. In contrast, negative-price periods were associated with much higher wind output, very low gas generation and a strong presence of negative imbalance.

In 2023, the transition from normal prices to upper-tail prices was especially clear. In the £0–100/MWh band, average gas generation was around 9.4 GW, and average wind output was around 7.2 GW. In the £250–300/MWh band, average gas generation increased to around 18.8 GW, while average wind output fell to around 5.5 GW. In the £300/MWh and above band, average gas generation increased further to around 20.0 GW and wind output remained low at around 5.3 GW. The share of low-wind periods below 8 GW also increased

to around 90% in the £300/MWh and above band. This suggests that the most extreme 2023 prices formed under a clear scarcity fingerprint.

The 2024 results show a similar but more compressed pattern. Most of the year remained in lower price bands, but when prices did reach the upper-tail bands, the physical conditions became more extreme. In the £300/MWh and above band, average wind output was only around 1.7 GW, while average gas generation was around 24.8 GW. Every period in this price band had wind below 8 GW and gas generation above 15 GW. This supports the interpretation that 2024 did not have frequent upper-tail stress, but the few high-price periods that did occur were physically severe.

In 2025, the price-band structure became more mixed. The year had both a major positive price shock and a high number of negative-price periods. In the £300/MWh and above band, average gas generation was around 22.8 GW, while average wind output was around 3.0 GW. All periods in this band had wind below 8 GW, and around 85% had gas generation above 15 GW. At the same time, the negative-price band had average wind output of around 11.9 GW and average gas generation of only around 3.2 GW, with negative imbalance present in around 87% of periods. This shows that 2025 contained both upper-tail scarcity behaviour and lower-tail surplus behaviour.

Overall, the price-band comparison shows that price levels are not only financial outcomes; they also reflect changing system conditions. As prices move into higher bands, the market becomes more strongly associated with low wind, high gas generation, positive imbalance and timing effects. As prices move into negative territory, the fingerprint reverses towards high wind, low gas generation and negative imbalance. This strengthens the paper's regime-based interpretation because the same drivers do not act in isolation; their importance changes depending on the price band and wider system state.

Price-Band Driver Fingerprint: Average Wind and Gas Generation, 2023–2025

● Average Wind ● Average Gas

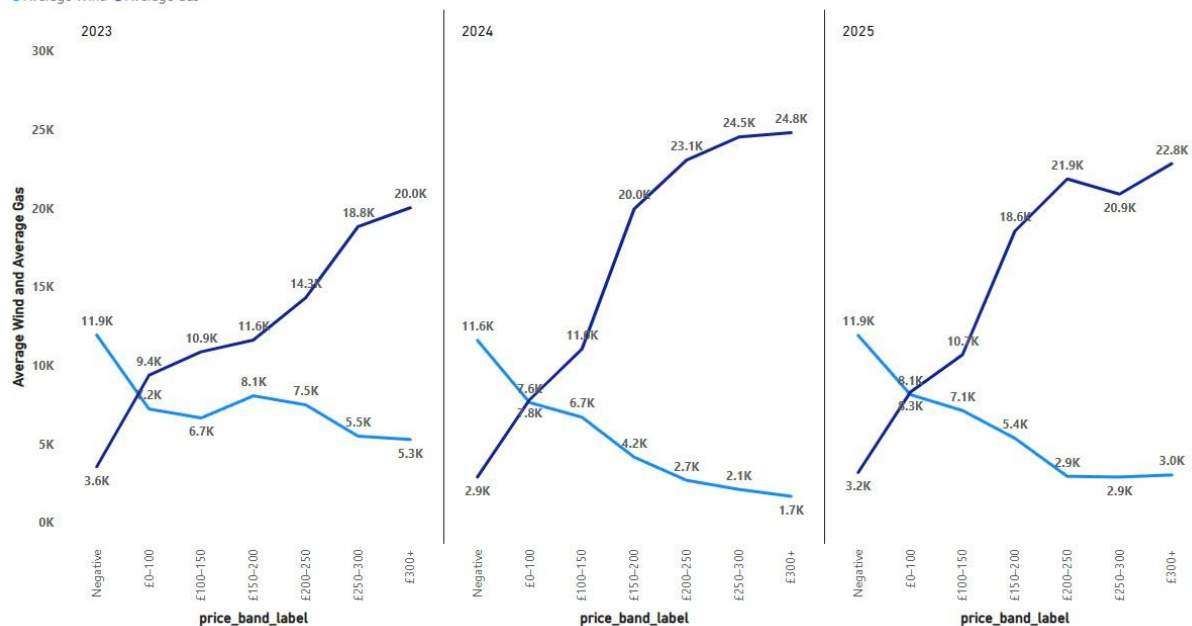


Figure 4.3 Price-band driver fingerprint across average wind and gas generation, 2023–2025.

The figure compares average wind and gas generation across price bands. Higher positive price bands are generally associated with increasing gas generation and lower wind output, while negative-price periods show the opposite pattern, with higher wind output and lower gas generation.

Year	Price band	Periods	Avg Price (£/MWh)	Avg Imbalance	Wind < 8 GW share	Gas > 15 GW share	Evening Share
2023	Negative	849	-31.69	-448.49	0.12	0.00	0.04
2024	Negative	1073	-18.97	-415.94	0.16	0.00	0.01
2025	Negative	1121	-16.42	-506.03	0.15	0.00	0.04
2024	£0–100	12196	62.69	-75.89	0.57	0.10	0.13
2025	£0–100	10803	66.55	-125.45	0.53	0.12	0.12
2023	£0–100	8443	57.82	-185.07	0.61	0.14	0.12
2023	£100–150	5342	122.34	151.37	0.66	0.20	0.21
2024	£100–150	4144	114.40	229.61	0.65	0.23	0.30
2025	£100–150	5128	117.19	214.78	0.62	0.24	0.27
2023	£150–200	2163	173.74	237.29	0.55	0.30	0.24
2023	£200–250	595	215.94	300.19	0.59	0.55	0.30
2023	£250–300	99	264.57	339.02	0.82	0.74	0.66
2023	£300+	21	501.97	294.89	0.90	0.76	0.86
2025	£150–200	400	163.15	289.97	0.82	0.76	0.47
2025	£250–300	15	272.53	261.66	1.00	0.80	0.67
2025	£300+	27	1,296.06	289.44	1.00	0.85	0.63
2024	£150–200	122	166.55	216.78	0.85	0.89	0.56
2024	£200–250	9	225.04	198.73	1.00	0.89	0.33
2025	£200–250	26	219.10	387.28	1.00	0.92	0.54
2024	£250–300	7	277.61	80.31	1.00	1.00	0.43
2024	£300+	17	452.20	333.66	1.00	1.00	0.47

Table 4.3 Price-band driver comparison, 2023–2025

The table summarises the number of half-hourly periods, average price, average imbalance and selected driver shares across each price band. It complements Figure 4.3 by showing how imbalance direction, low-wind exposure, high-gas exposure and evening timing change as prices move from negative and normal levels into upper-tail bands.

4.4 Threshold robustness and upper-tail sensitivity

The price-band analysis shows how market drivers change across different price levels. The next step is to test whether the same interpretation holds when the definition of a price spike is changed. This is important because any study of extreme prices depends partly on the threshold used to define an event. A £100/MWh period may reflect elevated market conditions, but it is not the same as a £250/MWh or £300/MWh event. For this reason, the analysis tests multiple thresholds: £100/MWh, £150/MWh, £200/MWh, £250/MWh and £300/MWh.

The threshold results show that the market fingerprint becomes more physically concentrated as the threshold increases. At lower thresholds, the sample includes a broad mix of elevated-price periods, including periods that may reflect normal tightness or moderate price pressure. At higher thresholds, especially £250/MWh and £300/MWh, the number of periods becomes much smaller, but the associated system conditions become more clearly linked to low wind output, high gas generation and evening timing.

In 2023, the number of periods above £100/MWh was large, with 8,220 half-hourly periods meeting this threshold. However, only 120 periods were above £250/MWh and only 21 periods were above £300/MWh. As the threshold increased, the scarcity fingerprint became more visible. The share of periods with wind below 8 GW increased from around 63% above £100/MWh to around 83% above £250/MWh and around 90% above £300/MWh. The share of periods with gas generation above 15 GW also increased from around 26% above £100/MWh to around 74% above £250/MWh and around 76% above £300/MWh. Evening timing became more important at the highest threshold, reaching around 86% of periods above £300/MWh.

The 2024 results show a different annual pattern but a similar upper-tail fingerprint. The number of periods above £250/MWh and £300/MWh was much smaller than in 2023, confirming that 2024 had less frequent upper-tail stress. However, when prices did reach these higher thresholds, the physical conditions were severe. All periods above £250/MWh and £300/MWh had wind below 8 GW and gas generation above 15 GW. This supports the view that 2024 was not a year of widespread spike activity, but its high-price events were strongly linked to physical scarcity conditions.

In 2025, threshold sensitivity became especially important because the year contained both a large January price shock and later mixed-tail behaviour. There were 5,596 periods above £100/MWh, but only 42 periods above £250/MWh and 27 periods above £300/MWh. At the £300/MWh threshold, all periods had wind below 8 GW, around 85% had gas generation above 15 GW, and around 63% occurred during the evening peak. This confirms that the most extreme positive prices in 2025 were associated with a tight scarcity fingerprint rather than general price elevation.

Overall, the threshold robustness analysis supports the main interpretation of the paper. The findings do not depend only on a single spike definition. As the price threshold rises, the same broad pattern becomes clearer: upper-tail prices are increasingly associated with low wind, high gas generation, positive imbalance and timing effects. This provides stronger evidence that extreme positive prices are not random outliers, but regime-dependent outcomes that emerge when several system conditions align.

Threshold Robustness: Low-Wind, High-Gas and Evening Shares, 2023–2025

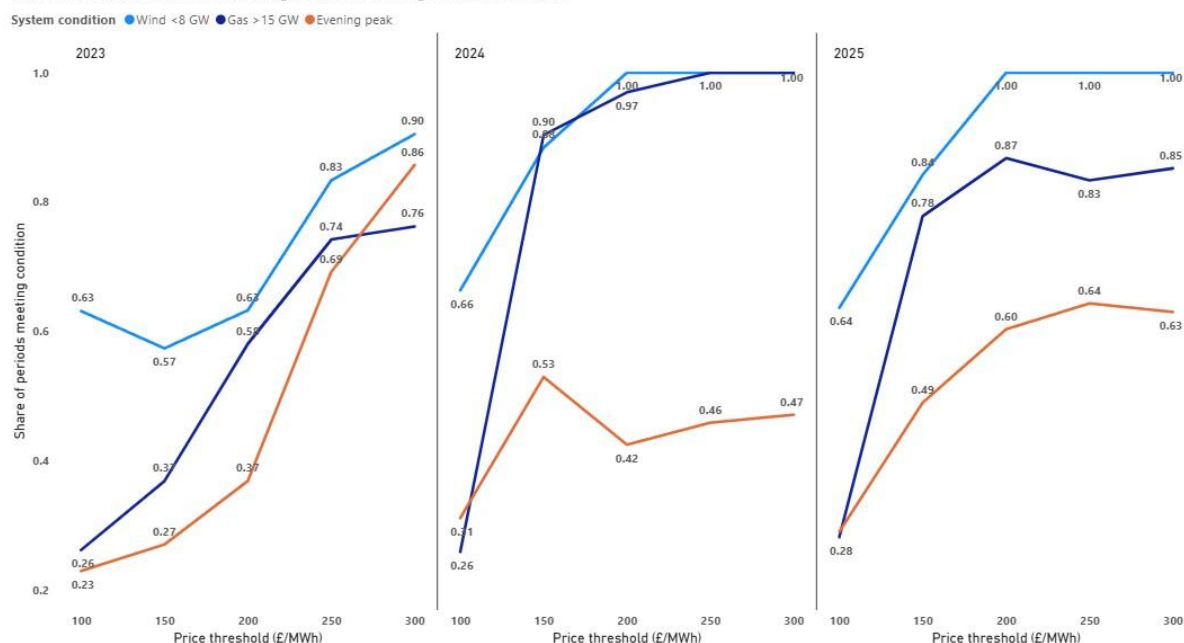


Figure 4.4 Threshold robustness of low-wind, high-gas and evening-peak shares, 2023–2025

Figure 4.4. Threshold robustness of low-wind, high-gas and evening-peak shares, 2023–2025.

The figure compares how selected system-condition shares change as the price threshold increases from £100/MWh to £300/MWh. The rising share of low-wind and high-gas periods at higher thresholds indicates that upper-tail prices are increasingly associated with a scarcity-led market fingerprint.

Year	Threshold (£/MWh)	Periods above threshold	Share of year	Avg price (£/MWh)	Gas > 15 GW share	Evening share	Wind <8 GW share
2023	100	8220	0.47	145.32	0.26	0.23	0.63
2023	150	2878	0.16	187.99	0.37	0.27	0.57
2023	200	715	0.04	231.07	0.58	0.37	0.63
2023	250	120	0.01	306.11	0.74	0.69	0.83
2023	300	21	0.00	501.97	0.76	0.86	0.90
2024	100	4299	0.24	117.71	0.26	0.31	0.66
2024	150	155	0.01	206.29	0.90	0.53	0.88
2024	200	33	0.00	353.21	0.97	0.42	1.00
2024	250	24	0.00	401.28	1.00	0.46	1.00
2024	300	17	0.00	452.20	1.00	0.47	1.00
2025	100	5596	0.32	127.05	0.28	0.29	0.64
2025	150	468	0.03	235.13	0.78	0.49	0.84
2025	200	68	0.00	658.50	0.87	0.60	1.00
2025	250	42	0.00	930.51	0.83	0.64	1.00
2025	300	27	0.00	1,296.06	0.85	0.63	1.00

Table 4.4 Compact threshold robustness summary, 2023–2025

Table 4.4 provides the numerical support for Figure 4.4. The table should be compact and should not include every column from the source dataset. Its purpose is to summarise the number of periods above each threshold, the average price within those threshold periods, and the main driver shares that define the upper-tail fingerprint.

4.5 Conditional spike probability analysis

The threshold and price-band results show that upper-tail prices are associated with low wind, high gas generation, imbalance and timing effects. However, those results do not fully show whether each driver is important on its own or whether spike risk increases when drivers occur together. To test this, the analysis uses a conditional spike-probability framework focused on the Q1 2023 stress regime, where the largest concentration of 2023 upper-tail price events occurred.

The results show that individual stress indicators had only limited explanatory power when considered separately. Gas generation above 15 GW was associated with a spike probability of around 6.28%, while positive imbalance above 150 MW was associated with a similar probability of around 6.07%. Wind below 8 GW had a lower singlecondition probability of around 4.81%. These results suggest that individual drivers were relevant, but not sufficient on their own to explain the periods of highest spike risk.

The combined-condition results show a clearer escalation. When positive imbalance above 150 MW coincided with gas generation above 15 GW, the spike probability increased to around 13.78%. When positive imbalance coincided with the evening peak window, the probability increased to around 17.45%. The strongest two-condition result was found when wind below 8 GW coincided with the evening peak window, where the spike probability reached around 19.81%. This indicates that timing effects were important, especially when combined with physical scarcity indicators.

The triple-condition results provide the clearest evidence for regime-based price formation. When positive imbalance above 150 MW, wind below 8 GW and the 16:00– 19:00 evening window occurred together, 47 spike periods were observed out of 138 qualifying periods. This produced a conditional spike probability of around 34.06%. This is substantially higher than the single-condition probabilities and supports the view that extreme prices are more likely when several system pressures align.

The conditional probability results therefore strengthen the paper's central argument. Price spikes are not explained well by one driver in isolation. Instead, spike risk increases as physical scarcity, imbalance direction and time-of-day effects combine. This finding also helps explain why some low-wind or high-gas periods do not produce extreme prices: the wider system regime must also be supportive of price stress.

Conditional Spike Probability Escalation in Q1 2023

Condition level ● Combined ● Single ● Triple

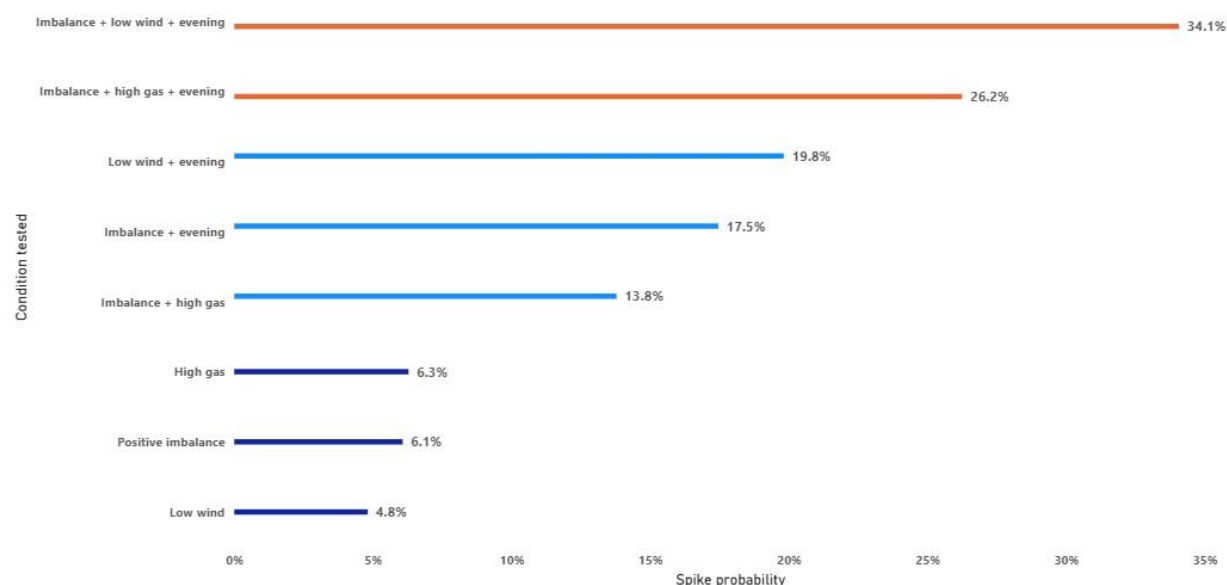


Figure 4.5 Conditional spike probability escalation in Q1 2023

The figure compares selected single-condition, combined-condition and triplecondition spike probabilities. Single drivers such as low wind, high gas generation and positive imbalance show relatively low spike probabilities, while the probability increases when these conditions occur together. The highest spike probability is observed when positive imbalance, low wind and evening timing coincide.

Condition level	Condition	Sum of Periods meeting condition	Sum of Spike periods	Sum of Spike probability
Single	Low wind	1892	91	4.81%
Single	High gas	1306	82	6.28%
Single	Positive imbalance	1301	79	6.07%
Combined	Imbalance + high gas	392	54	13.78%
Combined	Imbalance + evening	321	56	17.45%
Combined	Low wind + evening	318	63	19.81%
Triple	Imbalance + high gas + evening	141	37	26.24%
Triple	Imbalance + low wind + evening	138	47	34.06%

Table 4.4 Compact threshold robustness summary, 2023–2025

The table summarises selected conditional spike probabilities from the Q1 2023 stress regime. It shows that individual drivers had relatively low spike probabilities on their own, while the probability increased when imbalance, wind, gas and evening timing conditions occurred together.

5. Case Study and REMIT Cross-Check Analysis

5.1 Case Study selection and purpose

The previous section showed that extreme price behaviour in the GB power market was not evenly distributed across the 2023–2025 period. Price stress was concentrated in specific regimes, and the probability of a spike increased when several system conditions occurred together. To test whether the same pattern was visible during actual event windows, this section develops a case-study analysis of selected uppertail and lower-tail price events.

The case studies were selected to represent the main price regimes identified in the wider dataset. The purpose was not simply to list the highest prices in each year. Instead, each case was chosen because it represented a distinct form of price formation: Q1 compound scarcity in 2023, late-year physical scarcity in 2024, and mixed upper-tail and lower-tail behaviour in 2025. This allows the case-study analysis to connect the statistical findings from the previous section with specific market events.

The final case-study set contains sixteen cases across the three-year period. The 2023 case is used as the historical benchmark for Q1 compound scarcity, where the maximum observed price reached £1,950/MWh. The 2024 cases capture the late-year return of price stress, including the October transition period and December physical scarcity events. The 2025 cases are more detailed because that year contained both the largest positive price event in the dataset and several negative-price clusters. These cases include the January 2025 extreme spike, further January scarcity events, June and October positive spike cases, a July near-spike case, and lower-tail negative-price cases in March, April, June and September.

For each case, the analysis records the event window, maximum and minimum price, average price, dominant time band, regime group, average imbalance, average wind generation, average gas generation and average interconnector flow. These variables are used to describe the internal market fingerprint of each event. The same structure is applied across the selected 2025 cases so that positive spike events and negative-price clusters can be compared using a consistent method.

The 2025 cases are then cross-checked against REMIT notifications. This is done to assess whether material asset-availability events were present around the same time as the selected price windows. However, the REMIT layer is interpreted carefully. The analysis does not assume that every outage near a price spike caused the price movement. Instead, REMIT notifications are used as external context, helping to distinguish between repeated background asset-availability conditions and more casespecific availability evidence.

The aim of this section is therefore to move from aggregate patterns to event-level interpretation. If the case studies show the same fingerprints identified earlier, then the regime-based explanation becomes stronger. In particular, the section tests whether upper-tail events are associated with scarcity-led conditions such as low wind, high gas generation,

positive imbalance and timing effects, and whether lower-tail events are associated with surplus conditions such as high wind, negative imbalance and reduced thermal requirement.

Table 5.1 summarises the selected case studies used for the event-level analysis. The cases were chosen to represent the main regimes identified in the previous section, including Q1 2023 compound scarcity, late-year 2024 physical scarcity, and the mixed upper-tail and lower-tail structure observed in 2025. The table is intended as a compact event map rather than a full event log; detailed event windows and REMIT classifications are reported separately.

Year	Case ID	Max Price	Dominant time band	Regime	Price Behaviour
2025	2025_case_01_jan_08_extreme_spike	2,900.00	midday	jan_spike_stress	Extreme positive spike / upper-tail discontinuity
2025	2025_case_02_jan_20_evening_scarcity	678.48	evening_peak	jan_spike_stress	Positive spike event / upper-tail price stress
2025	2025_case_03_jan_22_morning_ramp_scarcity	450.00	morning_ramp	jan_spike_stress	Positive spike event / upper-tail price stress
2025	2025_case_04_jun_30_mixed_tail_spike	423.85	evening_peak	jun_mixed_tail	Positive spike event / upper-tail price stress
2025	2025_case_05_jul_01_near_spike	263.58	evening_peak	jul_sep_surplus	Near-spike / partial upper-tail conversion
2025	2025_case_06_oct_13_autumn_spike	487.00	midday	oct_mixed_transition	Positive spike event / upper-tail price stress
2025	2025_case_07_oct_14_evening_spike	300.00	evening_peak	oct_mixed_transition	Positive spike event / upper-tail price stress
2025	2025_case_08_oct_22_evening_spike	315.10	evening_peak	oct_mixed_transition	Positive spike event / upper-tail price stress
2025	2025_case_09_mar_30_negative_price_cluster	-0.46	night	feb_mar_broad_stress	Negative-price cluster / lower-tail event
2025	2025_case_10_apr_05_negative_price_cluster	105.00	afternoon	apr_may_surplus_shift	Negative-price cluster / lower-tail event
2025	2025_case_11_jun_22_negative_price_cluster	-1.23	afternoon	jun_mixed_tail	Negative-price cluster / lower-tail event
2025	2025_case_12_sep_06_negative_price_cluster	-0.75	late_evening	jul_sep_surplus	Negative-price cluster / lower-tail event
2023	Case 1	1,950.00	evening_peak	q1_stress	Major positive spike event
2024	Case 2	669.21	evening_peak	oct_nov_transition	Positive spike event
2024	Case 3	590.45	midday	dec_stress	Positive spike event
2024	Case 4	521.09	evening_peak	dec_stress	Positive spike event

Table 5.1 Selected case-study summary, 2023-2025

The selected cases show that price stress appeared in different forms across the study period. The 2023 and 2024 cases provide historical benchmarks for upper-tail scarcity, while the 2025 cases capture a broader market structure containing January scarcity, summer mixed-tail behaviour, October transition stress and several lower-tail negative-price clusters. This case selection therefore allows the paper to compare positive spike formation and negative-price formation within the same regime-based framework.

5.2 Positive spike case studies

The positive spike cases show that upper-tail price events appeared under different annual structures, but the underlying interpretation remained consistent: the highest prices occurred when system conditions moved towards scarcity or tighter balancing conditions. The selected cases cover Q1 2023 compound scarcity, late-year 2024 physical scarcity and the renewed upper-tail stress observed in 2025.

The 2023 benchmark case reached a maximum system price of £1,950/MWh during the Q1 stress regime. This event is important because it anchors the earlier finding that 2023 upper-tail behaviour was concentrated in Q1 rather than spread evenly across the year. The case is interpreted as a compound scarcity event, where price stress reflected the interaction

between Q1 system tightness, evening timing and upper-tail market behaviour rather than one isolated driver.

The 2024 cases show a different seasonal pattern. Unlike 2023, Q1 2024 did not produce the main price stress. Instead, upper-tail behaviour returned later in the year. The October transition case reached £669.21/MWh and represents the re-emergence of late-year price stress. The December cases reached £590.45/MWh and £521.09/MWh and are interpreted as physical scarcity events linked to low wind, high gas generation and December system tightness. These cases support the wider regime-level result that 2024 stress was more selective and late-year focused.

The 2025 positive spike cases show a more severe and mixed upper-tail structure. The January 8 case reached £2,900/MWh, the highest observed system price in the full 2023–2025 case-study set. This event is treated as the main 2025 upper-tail case because it combines an extreme price discontinuity with the January spike-stress regime. Additional January cases on 20 January and 22 January show that January 2025 did not contain only one isolated spike. Instead, it contained repeated scarcity episodes, including both evening scarcity and morning-ramp scarcity behaviour.

Later 2025 positive cases show that spike risk was not limited to winter. The June 30 case reached £423.85/MWh within the June mixed-tail regime and was associated with low wind, very positive imbalance and weaker interconnector support. This is important because the same month also contained negative-price exposure, showing that 2025 had mixed-tail behaviour rather than a simple winter-scarcity pattern. The October cases also show renewed upper-tail stress during the autumn transition regime, with prices reaching £487/MWh, £300/MWh and £315.10/MWh across selected windows.

Taken together, the positive spike cases support the regime-based interpretation developed earlier in the paper. The timing of upper-tail stress changed across years: Q1 in 2023, late-year in 2024, and January plus selected summer/autumn events in 2025. However, the common feature across these cases is that upper-tail prices were associated with scarcity-type conditions, system tightness or sudden price movement requiring both internal and external context. This reinforces the view that extreme positive prices are better explained by regime conditions than by individual variables alone.

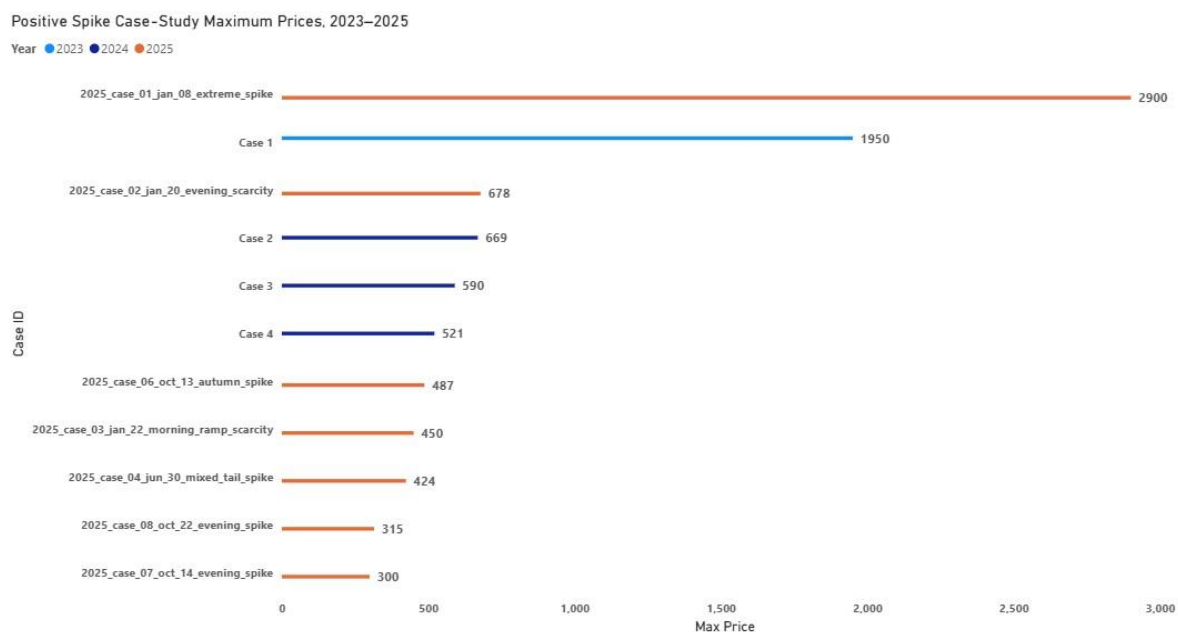


Figure 5.2 Positive spike case-study maximum prices, 2023–2025.

The figure compares maximum system prices across selected positive spike and nearspike case studies. It highlights the January 2025 extreme price discontinuity alongside the 2023 Q1 benchmark and the late-year 2024 scarcity cases.

5.3 Negative-price case studies

The negative-price case studies provide the lower-tail comparison to the positive spike cases discussed above. These events are important because they show that extreme price formation in the GB market is not limited to scarcity and high-price outcomes. The same regime-based framework can also be used to explain periods where system prices move sharply downwards.

The selected 2025 negative-price cases occurred across March, April, June and September. These cases were not treated as simple mirror images of positive price spikes. Instead, they were interpreted as lower-tail events associated with surplus or downward-price conditions. In these windows, the internal market fingerprint generally included high wind output, negative imbalance, lower gas generation and, in some cases, weaker interconnector support.

The March 2025 negative-price cluster occurred within the February–March broad stress regime and reached a minimum price of -£95/MWh. This case was associated with high wind output, negative imbalance and lower interconnector support. It provides an important example of how high renewable output and system length can push prices into the lower tail, even during a broader period that also contained elevated-price behaviour.

The April 2025 negative-price case occurred during the April–May surplus-shift regime. This period is interpreted as a transition into more surplus-sensitive market conditions, where reduced thermal requirement and negative imbalance increased exposure to lower prices.

Although the maximum price in the selected window was positive, the event is classified as a negative-price cluster because the wider window contained lower-tail behaviour and reflected downward-price formation.

The June 2025 negative-price case is particularly useful because June also contained a positive spike case. This supports the interpretation that 2025 was a mixed-tail year. Within the same broad month, the market could produce both positive spike risk and negative-price exposure depending on the direction of imbalance, wind output and system requirement. The June negative-price case was linked to high wind, negative imbalance and lower interconnector support, while the June positive case was linked to low wind, very positive imbalance and weaker interconnector support.

The September 2025 negative-price cluster occurred within the July–September surplus regime and reached a lower-tail price of around -£87/MWh. This case had a clear surplus fingerprint, with high wind output, negative imbalance and lower interconnector support. It reinforces the finding that the July–September period was more surplus-sensitive than scarcity-led, although the same period also contained a near-spike event in early July.

Taken together, the negative-price cases show that lower-tail price formation has a different system fingerprint from upper-tail price formation. Positive spike events were more closely linked to scarcity-type conditions such as low wind, high gas generation, positive imbalance and timing effects. Negative-price events were more closely linked to surplus-type conditions, including high wind, negative imbalance and lower thermal requirement. This supports the wider argument that extreme prices should be separated into upper-tail and lower-tail regimes rather than treated as one general category of price volatility.

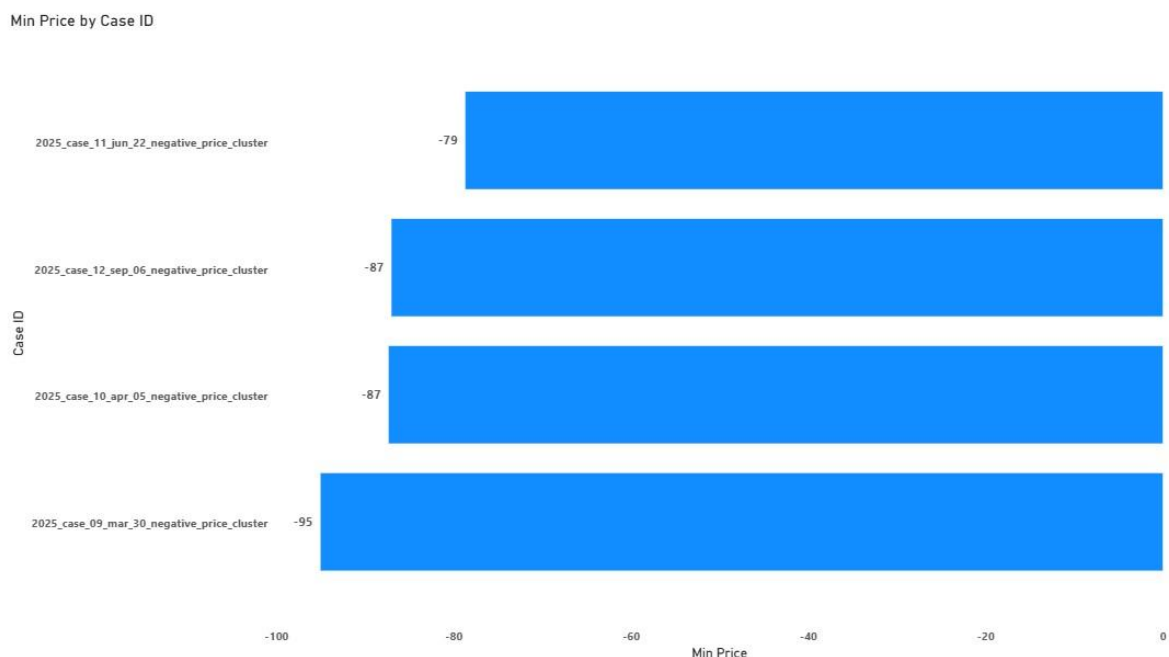


Figure 5.3 Negative-price case-study minimum prices, 2025.

The figure compares the minimum system prices across selected 2025 negative-price case studies. These lower-tail cases are associated with surplus-sensitive conditions, including high wind output, negative imbalance and reduced thermal generation requirement.

5.4 REMIT cross-check interpretation

The final stage of the case-study analysis applies a REMIT cross-check to the selected 2025 event windows (Elexon Limited, 2026). The purpose of this step is to compare the internal market fingerprint of each case with external asset-availability information. System price, wind generation, gas generation, imbalance and interconnector flows describe the internal market state, but they do not show whether generation, consumption or transmission availability events were present around the same period.

REMIT notifications provide this additional layer of market transparency.

For each 2025 case window, REMIT notifications were checked around the relevant event period. The analysis focused on material availability events, including production unavailability, consumption unavailability and transmission unavailability. These notifications were then interpreted alongside the internal price-event fingerprint. This approach allows the study to assess whether external availability conditions were present during price stress, without assuming that every notification directly caused the observed price movement.

A key finding from the REMIT overlay is that most 2025 cases were better interpreted as having **repeated background external context** rather than simple case-specific outage causality. Several assets appeared repeatedly across multiple case windows, meaning that their presence near an event was not enough to explain one individual price movement on its own. In these cases, REMIT evidence is treated as part of the wider availability background rather than as proof of a single-outage price effect.

This distinction is important for the interpretation of the January 2025 scarcity cases. The January 8 extreme spike reached £2,900/MWh, but the REMIT overlay does not imply that one asset outage alone caused the event. Instead, the case is interpreted as an internally stressed market regime, with REMIT availability conditions acting as external reinforcement. The dominant repeated REMIT assets identified around this case included NGIFA, SSETGSL, KGL, BEGL001 and DRAX, while NGVLL appeared as a more case-specific asset. This supports a regime-led interpretation: the price event occurred in a stressed system state, and asset-availability conditions formed part of the wider context.

The same logic applies to the October 2025 positive spike cases. These cases occurred during the autumn mixed-transition regime and were classified as repeated background external context. REMIT evidence showed repeated asset-availability conditions, including assets such as NGIFA, CENLANG, ICHPLLP, BEGL001, DRAX and CECL, with ELECL1 appearing as case-specific in some windows. This suggests that availability conditions may have reinforced market stress, but the price formation still needs to be understood through the wider regime: low wind, high gas generation, imbalance direction, interconnector position and time of day.

For the negative-price cases, REMIT evidence is interpreted even more carefully. Lowertail price formation is mainly linked to surplus-type internal conditions, such as high wind output, negative imbalance and lower thermal requirement. REMIT notifications may still provide relevant context, but they do not replace the internal explanation. In these cases, the REMIT cross-check helps confirm that external availability conditions existed, while the primary lower-tail interpretation remains linked to surplus and balancing conditions.

Overall, the REMIT cross-check strengthens the paper by adding an external validation layer, but it also prevents overclaiming. The evidence suggests that REMIT notifications are useful for contextualising price events, especially when material asset unavailability is present. However, the repeated appearance of several assets across multiple windows means that REMIT should not be treated as a simple causal list of outages. The case-study evidence therefore supports the central argument of the paper: GB price tails are primarily regime-led, with REMIT notifications acting as external context rather than direct proof of one-outage causality.

Case ID	Remit context type	Dominant repeated assets	Case specific assets
2025_case_01_jan_08_extreme_spike	repeated_background_external_context	NGIFA, SSETGSL, KGL, BEGL001, DRAX	NGVLL
2025_case_02_jan_20_evening_scarcity	repeated_background_external_context	NGIFA, CENLANG, BEGL001, DRAX, INNOGY01, DEESIDE	
2025_case_03_jan_22_morning_ramp_scarcity	repeated_background_external_context	NGIFA, BEGL001, DRAX, DEESIDE, KGL, EECL	
2025_case_04_jun_30_mixed_tail_spike	repeated_background_external_context	NGIFA, KGL, SPGEN01, SEABANK, CECL, INNOGY01	
2025_case_05_jul_01_near_spike	repeated_background_external_context	KGL, SPGEN01, SEABANK, CECL, INNOGY01, DRAX	
2025_case_06_oct_13_autumn_spike	repeated_background_external_context	NGIFA, CENLANG, ICHPLLP, BEGL001, DRAX, CECL	ELECL1
2025_case_07_oct_14_evening_spike	repeated_background_external_context	NGIFA, CENLANG, ICHPLLP, BEGL001, DRAX, CECL	ELECL1
2025_case_08_oct_22_evening_spike	repeated_background_external_context	SSETGSL, ICHPLLP, BEGL001, DRAX, DEESIDE, KGL	
2025_case_09_mar_30_negative_price_cluster	repeated_background_external_context	INNOGY01, DRAX, BEGL001, DEESIDE, KGL, EECL	
2025_case_10_apr_05_negative_price_cluster	repeated_background_external_context	NGIFA, KGL, SEABANK, SPGEN01, DRAX, BEGL001	
2025_case_11_jun_22_negative_price_cluster	repeated_background_external_context	NGIFA, CENLANG, SPGEN01, SEABANK, KGL, CECL	SUTTBGE
2025_case_12_sep_06_negative_price_cluster	repeated_background_external_context	SSETGSL, CENLANG, ICHPLLP, KGL, SPGEN01, SEABANK	

Table 5.4 REMIT cross-check classification for selected 2025 case windows.

The table summarises how REMIT notifications were interpreted for the selected 2025 case windows. Repeated background external context indicates that material REMIT events were present, but that dominant assets appeared across multiple event windows and should therefore be interpreted as contextual availability evidence rather than direct one-outage causality.

5.5 Summary of case-study evidence

The case-study analysis supports the wider results presented in Section 4. The selected events show that extreme price formation in the GB power market was not random, nor was it explained by a single recurring driver. Instead, each case reflected the wider regime in which it occurred.

The positive spike cases were mainly linked to scarcity-type conditions. In 2023, the main benchmark event occurred within the Q1 compound scarcity regime. In 2024, positive spike behaviour appeared later in the year, particularly during the October transition and December physical scarcity regimes. In 2025, the strongest upper-tail event occurred in

January, but further positive spike and near-spike cases appeared in June, July, and October. This confirms that upper-tail price risk can reappear outside winter when imbalance, wind, gas, and interconnector conditions align.

The negative-price cases show a different lower-tail structure. These cases were mainly associated with surplus-sensitive regimes, especially where high wind output, negative imbalance and lower thermal requirement created downward price pressure. The 2025 negative-price cases therefore demonstrate that the same regime-based framework can explain both high-price scarcity events and low-price surplus events.

The REMIT overlay adds an important external cross-check. It shows that asset availability conditions were present around selected 2025 event windows, but the evidence should not be interpreted as simple one-outage causality. The repeated appearance of several assets across multiple windows suggests that REMIT notifications often represent background availability context rather than direct explanations for individual price movements. This makes the interpretation more conservative and more consistent with the paper's central argument.

Overall, the case-study evidence strengthens the regime-based explanation of GB price formation. System prices moved into the upper or lower tail when several conditions interacted: wind output, gas generation, imbalance direction, interconnector support, time of day and asset-availability context. The case studies therefore bridge the gap between the aggregate statistical results and individual market events.

6. Discussions

6.1 Regime-based interpretation of price formation

The results of this study show that extreme price formation in the GB power market is better understood through market regimes than through isolated explanatory variables. Across 2023, 2024 and 2025, price stress did not appear evenly across the year, nor did it follow the same seasonal pattern in each year. Instead, upper-tail and lower-tail events were concentrated in specific periods where physical system conditions, imbalance direction and market timing interacted.

The 2023 results show a clear Q1 compound scarcity structure. Upper-tail prices were concentrated in the first quarter, with the main benchmark event reaching £1,950/MWh. This suggests that early-2023 price stress was not simply the result of high gas generation or low wind output alone but reflected a broader stress regime in which several conditions aligned. The quieter April–September period in 2023 further supports this interpretation, because low wind periods still occurred but did not consistently convert into extreme prices.

The 2024 results show that the timing of stress changed. Unlike 2023, Q1 2024 did not produce major upper-tail price behaviour. Instead, price stress returned later in the year, particularly during the October transition and December physical scarcity regimes. This

shows why annual averages can be misleading. A year may appear relatively quiet overall, while still containing short periods of concentrated price stress.

The 2025 results show the most complex structure. The year contained the largest positive price event in the study period, with the January 2025 spike reaching £2,900/MWh. At the same time, 2025 also contained the highest number of negative price periods. This makes 2025 a mixed-tail year, where both scarcity-led upper-tail events and surplus-led lower-tail events appeared within the same annual market architecture.

Taken together, these findings support the central argument of the paper: extreme GB power prices are regime-dependent outcomes. Price spikes and negative prices should not be treated as random anomalies. They are more usefully interpreted as the result of changing system states, where wind output, gas generation, imbalance direction, interconnector support, timing and asset availability interact.

6.2 Why single-driver explanations are insufficient?

A key implication of the analysis is that single-driver explanations are too limited for understanding extreme price behaviour. Low wind, high gas generation, positive imbalance and weak interconnector support are all relevant, but none of them fully explains price spikes on its own. Their impact depends on the wider regime in which they occur.

This is shown most clearly in the Q1 2023 conditional probability results. Individual drivers had relatively low spike probabilities when considered separately. For example, high gas generation alone and positive imbalance alone were each associated with spike probabilities close to 6%, while low wind alone had a probability below 5%. These figures suggest that each driver contributes to stress, but none is sufficient by itself to explain upper-tail price formation.

The probability increased when drivers were combined. Low wind together with evening timing produced a much higher spike probability, and the strongest result appeared when positive imbalance, low wind and evening timing occurred together. This shows that the most important signal is not simply whether one stress indicator is present, but whether several stress indicators appear at the same time.

This finding has practical importance. A market-monitoring framework based only on one variable, such as low wind output or high gas generation, may produce too many false signals. Many low-wind periods do not lead to spikes, and many high-gas periods reflect normal system operation rather than extreme scarcity. A regime-based framework is more useful because it considers the combined state of the system.

The same logic applies to lower-tail prices. Negative prices are not explained only by high wind output. They are more likely when high wind coincides with negative imbalance, lower thermal requirement and surplus-sensitive regimes. This means that both positive and negative price extremes require a multi-condition interpretation.

6.3 Upper-tail and lower-tail price formation

The results also show that upper-tail and lower-tail prices should be treated as different forms of price stress. Although both are extreme outcomes, they are not produced by the same market conditions. Positive spikes are mainly associated with scarcity-type regimes, while negative prices are more closely associated with surplus-type regimes.

Upper-tail price formation was most visible when low wind output, higher gas generation, positive imbalance and timing effects occurred together. This pattern appeared in different ways across the three years. In 2023, upper-tail stress was concentrated in Q1, where compound scarcity conditions produced the main benchmark spike. In 2024, upper-tail stress was less frequent but returned during the October–December period, especially in the December physical scarcity regime. In 2025, January produced the strongest positive price discontinuity, while June and October showed that upper-tail risk could also appear outside the main winter period.

Lower-tail price formation followed a different structure. Negative-price periods were more closely linked to high wind output, negative imbalance and reduced thermal generation requirement. This was especially visible in 2025, where the market combined a major January positive spike with the highest number of negative-price periods across the three-year dataset. The negative-price case studies in March, April, June and September show that lower-tail behaviour was not random; it appeared in regimes where surplus conditions and downward balancing pressure were more likely.

This separation matters because treating all extreme prices as one general category can hide the underlying market mechanism. A positive spike and a negative-price cluster may both be “extreme”, but they have different system fingerprints. Positive spikes tend to reflect scarcity, tightness and upward price pressure. Negative prices tend to reflect surplus, system length and downward price pressure. The same variables can also change meaning depending on the regime. For example, wind output is a scarcity risk when it is low, but it can become a surplus risk when it is high during weak demand or negative imbalance conditions.

The findings therefore support a two-sided interpretation of price tails (Rafizadeh, 2026; Hirth, 2013). The GB market is exposed to both upper-tail scarcity risk and lower-tail surplus risk. As renewable penetration increases and the system becomes more variable, both sides of the price distribution become important for market monitoring, forecasting and flexibility valuation.

6.4 REMIT as contextual evidence, not direct causality

The REMIT overlay adds an important external dimension to the analysis, but its interpretation must be cautious. REMIT notifications provide evidence of generation, consumption and transmission availability events, yet the presence of a notification near a price event does not automatically prove direct causality. A plant outage or interconnector unavailability event may matter more during a tight scarcity regime than during a surplus or

low-demand regime. For this reason, REMIT evidence is most useful when interpreted alongside the internal system state.

The 2025 case-study analysis shows this clearly. Several REMIT assets appeared repeatedly across multiple case windows. This means that their presence around a price event should not be treated as a simple explanation for that specific event. Instead, repeated REMIT appearances are better understood as background availability context. They may have contributed to the general availability environment, but the final price outcome still depended on wind output, gas generation, imbalance direction, interconnector conditions and market timing.

This distinction is especially important for the January 2025 scarcity cases. The January 8 event reached £2,900/MWh, but the evidence does not support reducing the event to a single-outage explanation. The internal market regime was already stressed, and REMIT notifications acted as external reinforcement rather than a standalone cause. The same applies to the October 2025 mixed-transition cases, where REMIT evidence was present but repeated across several windows.

For negative-price cases, REMIT should be treated even more carefully. Lower-tail price formation is mainly linked to surplus conditions such as high wind output, negative imbalance and lower thermal requirement. REMIT notifications may still provide useful context, but they do not replace the internal explanation of downward price pressure.

The main value of REMIT in this paper is therefore not causal attribution in a narrow sense. Its value is as a cross-checking layer. It helps confirm whether asset-availability stress was present around selected price windows and whether the evidence was casespecific or repeated background context. This makes the interpretation more robust and avoids the common mistake of treating every outage near a spike as the cause of that spike.

Overall, the REMIT results support the wider regime-based framework. Asset availability matters, but its price effect depends on the market regime in which it occurs. REMIT notifications should therefore be used as contextual evidence alongside system prices, fuel mix, imbalance and interconnector data.

6.5 Implications for market monitoring, trading and flexibility assets

The findings have practical implications for market monitoring, trading analysis and the operation of flexible assets. If extreme prices are regime-dependent, then monitoring tools should not rely only on single indicators such as low wind, high gas generation or asset outage notifications. These indicators are useful, but their value increases when they are interpreted together.

For market monitoring, the results suggest that a regime-based alert framework would be more useful than a simple threshold dashboard. For example, a low-wind alert on its own may generate many false signals. However, a combined alert that captures low wind, positive imbalance, high gas generation, evening timing and weaker interconnector support would be

more closely aligned with the conditions observed during upper-tail price events. Similarly, lower-tail alerts should combine high wind output, negative imbalance, lower gas generation and surplus-sensitive regimes.

For trading-oriented analysis, the results highlight the importance of conditional context. A trader or analyst looking only at price history may identify previous spikes but may not understand why some stressed periods converted into extreme prices while others did not. The conditional probability results show that the probability of a spike increased when multiple drivers coincided. This means that trading models and market monitoring tools should consider combinations of variables rather than treating each driver independently.

The results are also relevant for battery storage and flexible demand assets. Batteries can create value in both upper-tail and lower-tail regimes, but the operational strategy may differ. During scarcity-led upper-tail periods, value may come from discharging, reducing import, providing system support or capturing high-price spreads. During surplus-led lower-tail periods, value may come from charging at very low or negative prices, absorbing excess renewable generation, and positioning for later price recovery. A regime-based framework can therefore help identify not only when prices are extreme, but also what type of flexibility response is most appropriate.

The REMIT findings are particularly relevant for operational decision-making. REMIT notifications should not be ignored, but they should also not be treated as automatic trading signals. The analysis shows that repeated asset-availability events may form part of the background market context rather than directly explaining a single price movement. A stronger approach is to combine REMIT information with internal system indicators such as wind output, gas generation, imbalance, interconnector flows and time-of-day effects.

Overall, the practical value of this research is that it provides a transparent framework for interpreting price stress. The approach can support market monitoring, risk analysis, trading research and flexibility optimisation. It is especially relevant as the GB power system becomes more variable, more renewable-dependent and more reliant on flexible assets to manage both scarcity and surplus conditions.

7. Limitations and Future Work

This study provides a regime-based interpretation of extreme price formation in the GB power market, but several limitations should be recognised. The analysis is observational and empirical, rather than a formal causal model. It identifies relationships between system prices, wind generation, gas generation, imbalance, interconnector flows and REMIT notifications, but it does not claim to prove direct causality in a strict econometric sense.

A second limitation is that the analysis uses historical half-hourly data. This allows the study to examine realised price behaviour, but it does not include forecast uncertainty, demand forecasts, wind forecast errors or intraday market expectations. In practice, market participants respond not only to realised system conditions, but also to expected conditions

before delivery. Future work could extend the framework by including forecast data and comparing expected system stress with realised imbalance price outcomes.

The REMIT analysis also has limitations. REMIT notifications are useful for identifying asset-availability context, but they are not always straightforward to interpret. A notification may overlap with a price event without being the direct cause of that event. This study therefore treats REMIT as a cross-checking layer rather than a direct causal attribution tool. Future work could develop a more detailed REMIT event-matching framework, including asset type, unavailable capacity, location, duration and overlap with price-event windows.

The regime classification used in this paper is deliberately transparent and interpretable. This is useful for explaining market behaviour, but it is less formal than clustering, econometric regime-switching models, or machine-learning classification. Future research could compare the rule-based regime approach used here with more formal statistical regime-detection methods. This would help evaluate whether the same scarcity, surplus and mixed-tail regimes emerge under alternative modelling approaches.

Another area for future work is trading and flexibility optimisation. The results suggest that batteries, demand response, and flexible assets could benefit from regime-aware strategies. For example, upper-tail scarcity regimes may create value through discharge or import reduction, while lower-tail surplus regimes may create value through charging at extremely low or negative prices. Future work could connect the regime framework developed in this paper with battery dispatch modelling, revenue stacking, imbalance exposure, and flexibility-market participation.

Finally, the analysis could be extended geographically and operationally. Interconnector flows were included, but the study did not model DNO-level constraints, substation-level capacity, local congestion, or locational flexibility markets. Future research could combine national market data with DNO network data, REMIT notifications, and local flexibility signals to understand how national price stress interacts with local grid constraints.

Overall, these limitations do not weaken the main contribution of the paper. Instead, they define the boundary of the study. The paper provides a transparent empirical framework for understanding how extreme GB prices form under different regimes, while future research can extend the work into forecasting, causal modelling, asset optimisation, and local network analysis.

8. Conclusion

This paper examined extreme price formation in the GB power market between 2023 and 2025 using half-hourly system price, fuel mix, imbalance, interconnector and REMIT notification data. The main objective was to understand whether extreme positive and negative prices are better explained by individual drivers or by wider market regimes.

The findings show that extreme prices are strongly regime dependent. Price stress was not evenly distributed across the three-year period. In 2023, upper-tail price behaviour was concentrated in a Q1 compound scarcity regime, with the main benchmark event reaching £1,950/MWh. In 2024, price stress was less frequent and appeared later in the year, particularly during the October transition and December physical scarcity regimes. In 2025, the market showed a mixed-tail structure, combining the largest observed positive price event of £2,900/MWh with the highest number of negative-price periods across the study window.

The analysis also shows that single-driver explanations are insufficient. Low wind, high gas generation, positive imbalance, weak interconnector support and asset availability all matter, but none of these variables fully explains extreme prices on its own. The conditional probability results show that spike probability increased when several conditions occurred together, especially when positive imbalance, low wind and evening timing coincided. This supports the view that price-tail formation is driven by interacting system conditions rather than one isolated market variable.

The price-band and threshold robustness analysis further supports this conclusion. As positive price thresholds increased from £100/MWh to £300/MWh, the associated market fingerprint became more strongly linked to low wind output, high gas generation and evening timing. Negative-price periods showed a different structure, being more closely associated with high wind output, negative imbalance and lower thermal generation requirement. This confirms that upper-tail and lower-tail prices should be treated as distinct forms of market stress.

The REMIT cross-check adds an external layer of evidence, but it also shows the need for careful interpretation. REMIT notifications were useful for identifying asset availability context around selected 2025 case windows. However, repeated asset appearances across multiple event windows suggest that REMIT evidence should not be treated as simple one-outage causality. Instead, REMIT is most useful as contextual evidence when combined with internal market indicators.

Overall, the paper contributes a transparent regime-based framework for analysing GB power price formation. The results suggest that market monitoring, trading analysis and flexibility optimisation should move beyond single-variable alerts and instead consider the combined state of wind output, gas generation, imbalance direction, interconnector support, time of day and asset-availability context. As the GB power system becomes more renewable-dependent and more variable, this type of regime-based approach can help explain both scarcity-led price spikes and surplus-led negative-price events.

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Appendix

1. **Appendix A:** Annual dataset validation summary
2. **Appendix B:** Full regime fingerprint table
3. **Appendix C:** Full price-band driver comparison
4. **Appendix D:** Full threshold robustness results
5. **Appendix E:** Full conditional probability tables
6. **Appendix F:** Full REMIT case-window classification